

# ABM BUSINESS LOGIC BIBLE

## §12 v2 — DISTRIBUTED REALITY COORDINATION SYSTEM

10 Iterations · 10 Architectural Layers · 28 New Rules · 4 Integration Decisions · 1 New OPEN-Q

§12 v1 specified field-level push/pull and reconciled it against INT-CRM's architecture. §12 v2 reframes the engine itself. **§12 is not a sync engine. It is a Distributed Reality Coordination System.** Every prior module asked 'what is true?' §12 asks 'how do two incomplete, delayed, partial observers of the same reality converge?' ABM sees signals, intent, influence, relationships, narratives. CRM sees meetings, deals, sales notes, negotiations. Neither sees everything. Authority over a field means final decision rights, not superior knowledge — and the purpose of sync is not agreement between two systems, it is convergence toward reality itself.

|            |    |                       |            |
|------------|----|-----------------------|------------|
| Iterations | 10 | Layers                | 10         |
| New rules  | 28 | Integration decisions | 4 explicit |

### FOUR INTEGRATION DECISIONS

**Authority Registry generalizes INT-CRM-02:** the existing ABM-owns-signals/CRM-owns-deal-stage split and §11 v2's Decision Ownership Boundary both register as INSTANCES of one general Authority Registry, not two ad-hoc rules sitting beside a new generic one.

**Change Provenance reuses the SIGNAL/SCORE append-only pattern** (V2 §2.3, SCORE-CALC-002) generalized to every CRM-synchronized field — not a new logging mechanism.

**Origin Metadata IS EPIS-SRC-01** (reliability ■ incentive-bias, separated) applied to CRM-synchronized facts specifically — not a parallel provenance tracker.

**Reality Convergence is measured concretely:** declining conflict rate on the same field over time — not a vibes-based 'getting closer to truth,' which would be unfalsifiable.

## THE TEN LAYERS — §12 v2 ARCHITECTURE MAP

| #  | Layer                               | From iter. | What it adds                                                                             |
|----|-------------------------------------|------------|------------------------------------------------------------------------------------------|
| 1  | Temporal vs Logical Conflict (Time) | 1          | Different timestamps converging on the same reality ≠ fundamentally contradictory claims |
| 2  | Partial Reality (Real)              | 2          | Authority ≠ omniscience; log what truth the losing side held                             |
| 3  | Human Update Latency (Lat)          | 3          | CRM truth is delayed truth; distinguish Unknown from Outdated                            |
| 4  | Reality Reconstruction (Fusion)     | 4          | A Unified Reality Timeline fuses signals + outreach + meetings + deals                   |
| 5  | Convergence (Epist)                 | 5          | Success = declining conflict rate over time, not sync completion                         |
| 6  | Ownership Conflict (Own)            | 6          | Authority Registry, runtime-enforced; generalizes INT-CRM-02 + §11 v2 HAND-ACC-01        |
| 7  | Event Ordering (Order)              | 7          | Event Time ≠ Arrival Time; state keys off Event Time — OPEN-Q on source reliability      |
| 8  | Auditability (Audit)                | 8          | Change Provenance = the existing append-only pattern, generalized                        |
| 9  | Data Lineage (Lineage)              | 9          | Origin Metadata = EPIS-SRC-01 applied to CRM-synchronized facts                          |
| 10 | Sync Economics (Econ)               | 10         | Materiality Threshold — sync only what changes human behavior                            |

## LAYER 1 — TEMPORAL vs LOGICAL CONFLICT (TIME)

ABM says HOT at 10:00. A rep marks Closed Lost in HubSpot at 10:01. Sync hasn't run yet at 10:02. Both systems were correct — at different times. Not every conflict is a truth conflict; many are simply two snapshots of a reality that changed between them.

| Rule ID     | Phase | Specification                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-TIME-01 | P1    | Every sync conflict is classified as <b>Temporal</b> (both values were true at their respective timestamps; the disagreement resolves itself once the later event is applied) or <b>Logical</b> (the values are fundamentally contradictory regardless of timing — e.g. ABM=HOT and CRM=CLOSED_LOST cannot both be simply 'true at different times' in a way that self-resolves). |
| CRM-TIME-02 | P1    | Temporal conflicts resolve automatically by applying the later Event Time's value (Layer 7) without escalation. Only Logical conflicts trigger INT-CRM-04's conflict-resolution escalation — this distinction is what prevents the unnecessary escalation volume that would otherwise drown the genuinely meaningful conflicts in noise.                                          |

## LAYER 2 — PARTIAL REALITY (REAL)

| Rule ID     | Phase | Specification                                                                                                                                                                                                                                                                                                                                          |
|-------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-REAL-01 | P1    | Authority (per the Layer 6 Registry) means final decision rights on a field's VALUE, not superior knowledge about the underlying reality. A field's Authority Owner can be wrong; the Registry resolves WHO decides, not WHAT is objectively true.                                                                                                     |
| CRM-REAL-02 | P1    | Every conflict resolution explicitly logs the <b>Lost Information</b> — the losing side's value and its supporting context — even though it does not win. CRM wins deal-stage conflicts (INT-CRM-04), but ABM's discarded signal-based context is retained as a Tier 6 input, not deleted; it may still be useful even when it didn't carry the field. |

### LAYER 3 — HUMAN UPDATE LATENCY (LAT)

CRM doesn't reflect reality — it reflects what a human entered, whenever they got around to entering it. A deal lost Monday and logged Friday means four days of stale ground truth that the engine was nonetheless treating as authoritative.

| Rule ID    | Phase | Specification                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-LAT-01 | P2    | Every CRM-sourced field carries a <b>Freshness Confidence</b> derived from: the field owner's historical update lag, the field's typical update frequency, and time since last change. This is distinct from the field's Authority (Layer 6) — CRM remains authoritative even when Freshness Confidence is low; low confidence flags the value as POSSIBLY stale, not WRONG.                                        |
| CRM-LAT-02 | P2    | Repeated delayed updates from the same owner or team accumulate into a <b>CRM Staleness Risk</b> score, feeding the same per-subsystem trust model as §8 v2 QC-HUMAN-08 and §11 v2 HAND-WF-07 — a rep whose CRM updates are chronically late is a degrading subsystem exactly like a degrading AI module, tracked the same way.                                                                                     |
| CRM-LAT-03 | P1    | The engine distinguishes <b>Unknown</b> (no value has ever been recorded) from <b>Outdated</b> (a value exists but Freshness Confidence is low) as genuinely different states — an Outdated deal stage still gates automation per INT-CRM-04, but the handoff package (§11 v2 HAND-AUTH-02) surfaces 'last confirmed N days ago' rather than presenting it with the same confidence as a value updated an hour ago. |

## LAYER 4 — REALITY RECONSTRUCTION (FUSION)

ABM stores signals. CRM stores sales actions. Neither alone explains what actually happened to an opportunity. Sync's deeper purpose is fusing both into one coherent history, not merely keeping two databases from disagreeing.

| Rule ID       | Phase | Specification                                                                                                                                                                                                                                                                                                    |
|---------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-FUSION-01 | P1    | Every synchronized account maintains a <b>Unified Reality Timeline</b>: signals (§3), outreach touches (§9), engagement events (§10), handoffs and Decision Records (§11 v2), and CRM-pulled deal events (§12.2), merged chronologically by Event Time (Layer 7), not by which system happened to record them.   |
| CRM-FUSION-02 | P1    | §13 Analytics consumes the Unified Reality Timeline as its primary input for opportunity-history analysis, NOT isolated per-system event logs — a KPI computed from ABM events alone or CRM events alone is, by construction, working from a partial reality (Layer 2) and is flagged as such in any §13 report. |

## LAYER 5 — CONVERGENCE (EPIST)

| Rule ID      | Phase | Specification                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-EPIST-01 | P1    | The success metric for §12 is <b>Reality Convergence</b>, operationalized concretely as: a DECLINING Logical-conflict rate (Layer 1) on the same field over time, not merely 'sync completed' or 'fields match.' Two systems that agree because one blindly overwrote the other have NOT converged; two systems that increasingly agree because the underlying ambiguity was resolved HAVE. |
| CRM-EPIST-02 | P2    | A field with a persistently elevated Logical-conflict rate (not declining) becomes a <b>Reality Investigation Opportunity</b> flagged to §8 v2 QC-SYS-01 as a Failure Cluster — persistent disagreement usually indicates a missing signal source, a miscalibrated authority assignment, or a process gap, not random noise.                                                                |

## LAYER 6 — OWNERSHIP CONFLICT THEORY (OWN)

INT-CRM-02 already split authority for the major fields V2 anticipated. But every NEW field added to the system — a 'Strategic Priority' property, a custom deal attribute — is a fresh opportunity for ambiguous ownership unless the discipline is generalized into a registry, not re-derived field by field.

| Rule ID    | Phase | Specification                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-OWN-01 | P1    | Every synchronized field has exactly ONE declared <b>Authority Owner</b>: ABM, CRM, Governance, Human, or Derived (computed from other authoritative fields, itself carrying its own authority). No field has shared or co-equal ownership, ever.                                                                                                                                                                            |
| CRM-OWN-02 | P1    | A field with no declared Authority Owner CANNOT enter synchronization — this is enforced at the point a new field is added to the sync configuration, not discovered later when a conflict occurs.                                                                                                                                                                                                                           |
| CRM-OWN-03 | P1    | The <b>Authority Registry</b> is a runtime-enforced artifact (not documentation): INT-CRM-02's existing ABM/CRM split and §11 v2 HAND-ACC-01's Decision Ownership Boundary (AI Recommendation / Human Decision / Human Execution) both register into it as existing entries, alongside every field added since. The Registry, not separate prose rules, is what §12's conflict-resolution logic actually queries at runtime. |

## LAYER 7 — EVENT ORDERING (ORDER)

Tier→HOT at 10:00, Meeting Booked at 10:01, Closed Lost at 10:02 — but network delay means the system RECEIVES them as Closed Lost, then HOT, then Meeting Booked. Keying state off arrival order instead of event order silently corrupts the account's state.

| Rule ID      | Phase | Specification                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-ORDER-01 | P1    | Every synchronized event stores BOTH an <b>Event Timestamp</b> (when it actually happened, per the source system) and an <b>Arrival Timestamp</b> (when the ABM engine received it) as separate fields, never collapsed into one.                                                                                                                                                      |
| CRM-ORDER-02 | P1    | State transitions are computed from Event Timestamp, never Arrival Timestamp. An event that arrives late but has an earlier Event Timestamp than the current state's last-applied event does NOT blindly overwrite — it triggers Layer 1's conflict classification against what's already been applied.                                                                                |
| CRM-ORDER-03 | P1    | A late-arriving event with an earlier Event Timestamp than already-applied events triggers <b>State Reconciliation</b> : the timeline (Layer 4) is recomputed with the correctly-ordered event inserted, and any state derived from the now-corrected sequence is re-evaluated — never a blind overwrite that could silently revert a correct later state to an incorrect earlier one. |

### OPEN-DESIGN-QUESTION OPEN-Q12.3 — EVENT TIMESTAMP RELIABILITY ACROSS SOURCES

**Question:** CRM-ORDER-01/02 assume every source can supply a trustworthy Event Timestamp. Not all sources are equal: HubSpot webhooks typically carry reliable event-time metadata; some signal sources (§3 INT-SIG) may only have a detection time, with the true underlying event time inferred or approximate. What's the minimum-viable ordering guarantee when Event Time itself carries uncertainty?

**Candidate options:** (a) Treat detection time as Event Time when no better timestamp exists — simplest, but reintroduces the arrival-time problem under a different name for those sources. (b) Carry an explicit Event Time Confidence alongside the timestamp (reusing EPIS-RCM's confidence pattern) and only trigger full State Reconciliation (CRM-ORDER-03) when confidence is high enough to justify it; low-confidence reordering logs a flag but doesn't reconstruct state. (c) Require every source to be explicitly classified as Event-Time-Reliable or Detection-Time-Only at integration time, with different handling per class.

**Status:** Genuinely unresolved — depends on per-source engineering reality (does HubSpot's webhook payload reliably carry the underlying event time, or only HubSpot's own record-update time?) that isn't fully knowable from the business-logic layer alone. Recommended path if forced to pick: (b), since it degrades gracefully rather than either ignoring the uncertainty or blocking integration of imperfect sources.

## LAYER 8 — AUDITABILITY (AUDIT)

Enterprise banking customers will ask why an account became DORMANT, who changed it, and based on what. This is not a new discipline — it's the append-only pattern already established for SIGNAL and ACCOUNT\_SCORE, generalized to every CRM-synchronized field.

| Rule ID      | Phase | Specification                                                                                                                                                                                                                                                                                   |
|--------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-AUDIT-01 | P1    | Every field change stores <b>Change Provenance</b> : previous value, new value, actor (which system/person), timestamp, and reason where available — using the SAME append-only discipline as V2 §2.3 (signals) and SCORE-CALC-002 (scores): never overwritten, always a new historical record. |
| CRM-AUDIT-02 | P1    | Every synchronized object supports <b>Reality Replay</b> : reconstructing the complete history of any field's values over time from its Change Provenance records, consumed by the Unified Reality Timeline (Layer 4) and available for compliance/audit requests.                              |
| CRM-AUDIT-03 | P1    | No destructive updates anywhere in the CRM sync layer — this generalizes the append-only principle the Bible has used since §2 (SIGNAL, ACCOUNT_SCORE) to its full scope: every synchronized field, not only the two V2 originally called out.                                                  |



## LAYER 9 — DATA LINEAGE (LINEAGE)

“Budget approved” could come from a CRM note, an SDR’s observation, a LinkedIn signal, a meeting outcome, or a human override — and fact quality depends heavily on which. This is EPIS-SRC-01’s reliability/incentive-bias separation, applied specifically to facts that flow through CRM sync.

| Rule ID        | Phase | Specification                                                                                                                                                                                                                                                                                              |
|----------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-LINEAGE-01 | P1    | Every synchronized fact carries <b>Origin Metadata</b> using EPIS-SRC-01’s existing reliability/incentive-bias axes (Human / AI / CRM / External Source / Partner each get their own reliability and bias profile) — not a new, separate provenance taxonomy.                                              |
| CRM-LINEAGE-02 | P2    | Every fact maintains a <b>Transformation Chain</b> : origin → any derived calculations performed on it → current usage. A score derived from a CRM-sourced fact carries that fact’s lineage forward, not just its current numeric value.                                                                   |
| CRM-LINEAGE-03 | P1    | Any model or scoring function (§4, §7 v2, §10 v2) that consumes a CRM-synchronized fact reads Fact + Lineage together, exactly as EPIS-RCM-01 already requires for every other fact in the architecture — CRM-sourced facts are not exempt from the RCM stamp just because they crossed a system boundary. |

## LAYER 10 — SYNC ECONOMICS (ECON)

A score moving 80.1→80.2→80.3→80.4 doesn't deserve four CRM pushes — that's noise, not signal, and every sync has a real cost (API calls, conflict-resolution risk, human attention spent reviewing a meaningless change). This is §6 v3's value-of-information discipline applied to outbound sync.

| Rule ID     | Phase | Specification                                                                                                                                                                                                                                                                                                                                  |
|-------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-ECON-01 | P1    | Every synchronized field has a <b>Materiality Threshold</b> below which a change does not trigger a push: e.g. a score change under 5 points does not push to HubSpot. The threshold is field-specific, not a single global cutoff (a tier CHANGE is always material; a score WOBBLE within a tier is not).                                    |
| CRM-ECON-02 | P1    | Sync is optimized for <b>Decision Impact</b> , not data freshness — reuses §6 v3 Layer 1's marginal-information-value discipline (ENR-COMP-02), applied to outbound propagation instead of inbound acquisition: push when the change would plausibly alter what a human does next, not whenever the underlying number ticks.                   |
| CRM-ECON-03 | P1    | The materiality test is explicitly: 'will this change alter human behavior?' A tier change, a deal-stage-relevant state transition, or a new Decision Record clears this test readily; a sub-threshold score fluctuation does not. Sub-threshold changes still accumulate in Change Provenance (Layer 8) — they are not lost, only not pushed. |

## §12 v2 RULE ROLL-UP & PLACEMENT STATUS

| Layer     | Rules              | Phase   | Cross-section binding                                          |
|-----------|--------------------|---------|----------------------------------------------------------------|
| 1 Time    | CRM-TIME-01..02    | P1      | INT-CRM-04 (escalation scoped to Logical only)                 |
| 2 Real    | CRM-REAL-01..02    | P1      | Tier 6 Strategic Memory (Lost Information retained)            |
| 3 Lat     | CRM-LAT-01..03     | P1 / P2 | §8 v2 QC-HUMAN-08, §11 v2 HAND-WF-07 (same trust model)        |
| 4 Fusion  | CRM-FUSION-01..02  | P1      | §3/§9/§10/§11 v2 as timeline inputs, §13 primary data source   |
| 5 Epist   | CRM-EPIST-01..02   | P1 / P2 | §8 v2 QC-SYS-01 Failure Cluster                                |
| 6 Own     | CRM-OWN-01..03     | P1      | Generalizes INT-CRM-02 + §11 v2 HAND-ACC-01 into one Registry  |
| 7 Order   | CRM-ORDER-01..03   | P1      | OPEN-Q12.3 (source timestamp reliability)                      |
| 8 Audit   | CRM-AUDIT-01..03   | P1      | V2 §2.3 + SCORE-CALC-002 (same pattern, generalized)           |
| 9 Lineage | CRM-LINEAGE-01..03 | P1 / P2 | EPIS-SRC-01 (reused), EPIS-RCM-01 (no exemption for CRM facts) |
| 10 Econ   | CRM-ECON-01..03    | P1      | §6 v3 ENR-COMP-02 (same discipline, outbound direction)        |

### §12 v2 COMPLETE — WHAT IT ESTABLISHED

§12 is no longer a sync engine. It is a **Distributed Reality Coordination System**: it distinguishes conflicts that resolve themselves from those that need escalation, recognizes authority as decision-rights rather than omniscience, accounts for human update latency as a first-class uncertainty, fuses ABM and CRM events into one reality timeline, measures success as declining disagreement over time rather than instantaneous agreement, generalizes field ownership into a runtime-enforced Registry, separates event time from arrival time to prevent silent state corruption, generalizes the existing append-only audit pattern to every field, applies EPIS's source-reliability model to CRM-sourced facts specifically, and syncs only what would actually change human behavior. 28 new rules, four integration decisions binding every new layer to existing architecture rather than parallel mechanisms, and one genuinely open question about event-timestamp reliability that depends on per-source engineering reality rather than business-logic preference.

§12 v2 is recommended-stopped here per the standard exhaustion assessment — the remaining frontier (multi-CRM support, disaster recovery, cross-region replication) is platform engineering, not ABM business logic.

### PLACEMENT MAP UPDATE

§12 status: Complete-with-Opens (OPEN-Q12.1, OPEN-Q12.2 from §12 v1 unchanged, plus new OPEN-Q12.3 on event-timestamp reliability).

### ON THE HORIZON

§13 Analytics & Feedback Loop Engine — the next genuinely large architectural territory per both assessments. §13 inherits the Unified Reality Timeline (Layer 4) as its primary data source, Reality Convergence (Layer 5) as a KPI candidate, and must make room for Trust Preservation Score, Combined Decision Quality, Cause-Attributed outcomes, and Reputation Yield alongside V2's original 9 KPIs.